

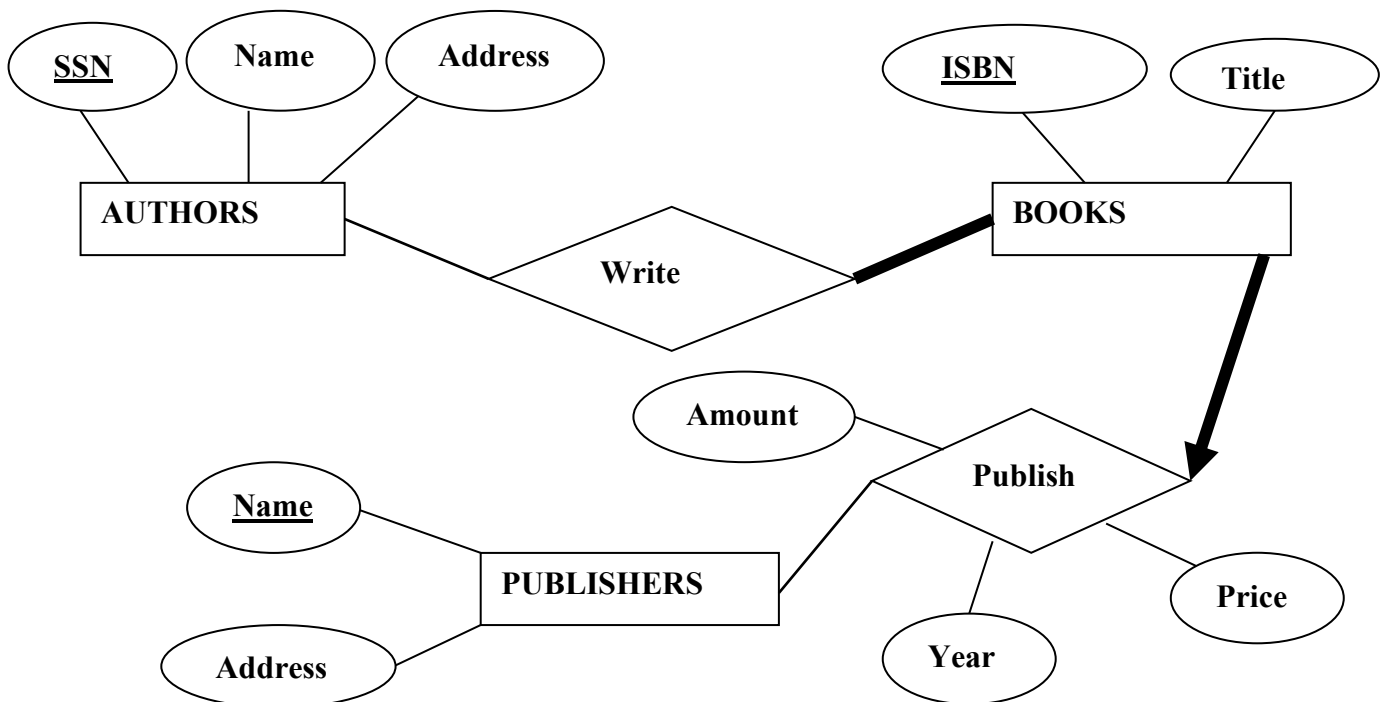
CEN350 Database Management Systems Final Homework

Due 15.06.2020 until 23:59

Solve the below questions by yourself (it is NOT A GROUP HOMEWORK) and write your results by using Word or any word processing software. Upload .doc, .docx, or .pdf document to the system. Handwritten homework will not be accepted.

QUESTIONS

1. (30 points) For the below E/R diagram, write SQL commands to describe the most suitable relational database schema which includes names of the tables, attributes, keys, and all constraints.



2. (25 points) In the below table customers buy some products in a given amount; and each product has barcode, name, weight, and price.

Customer ID	Customer Name	Bar Code	Product Name	Weight	Price (TL)	Amount bought
111	Aslı Kaya	112233	Milk	1 lt	5	2
111	Aslı Kaya	223344	Coffee	100 gr	6	1
122	Melis Çelik	112233	Milk	1 lt	5	4
133	Mesut Demir	223344	Coffee	100 gr	6	2

- List all the functional dependencies for the above table.
- What is the primary key of the above table? Why? Explain.
- Is the above table in BCNF? Explain your answer. If it is not in BCNF, convert it into BCNF.

3. (30 points) Consider the below relational database schema where a course can be offered by one or more teachers; and a student can enroll to any number of courses.

COURSES (CourseID, CourseName, Credit)

TEACHERS (SSN, Name)

OFFER (CourseID, SSN)

STUDENTS (StudentID, Name, Department)

ENROLL (StudentID, CourseID, Semester)

Write the following queries in standard SQL (you should use SQL-99 standard that is covered in your text book).

- Find the names of the students who enroll to courses that are enrolled by student whose name is “Mehmet Öz”.
 - Find the names of the teachers who offer courses that are enrolled by “Mehmet Öz”.
 - Find the names of courses that are enrolled by all “Computer Engineering” students.
 - Find the names of courses that are enrolled by some “Computer Engineering” students, and are not offered by “Aslı Kaya”.
 - Find the names of the courses that are enrolled by at least 5 “Computer Engineering” students.
4. (15 points) Consider the below schedule, where two transactions T1 and T2 run concurrently on the database objects A and B. In the below schedule “R” means Read operation, “W” means Write operation on the database object A or B.

T1	T2
R(A)	
R(B)	
	W(B)
	W(A)
W(B)	
R(A)	
commit	
	commit

- Explain all the problems in the above schedule.
- Re-write the above schedule according to Strict 2 Phase Locking concurrency control protocol.